

ICU mortality (SUPPORT2) — subgroup deployment-readiness audit

Model	Logistic regression (13 clinical features + disease class, cancer status)
Cohort	SUPPORT2, UCI id 880 (N=9,105); audited on held-out test partition n=1821
Outcome	In-hospital mortality (25.9% in the test partition)
Threshold	0.3
Predictions	Out of fold — fitted on a disjoint 60% training partition
Uncertainty	Patient-level bootstrap, B=2000, random_state=99
isitfair version	0.4.0

Audit summary

N: 1821	Events: 472	Prevalence: 0.259	Threshold: 0.300
Bootstrap resamples: 2000	Resampling: subgroup-stratified patient-level	Attributes: 3	

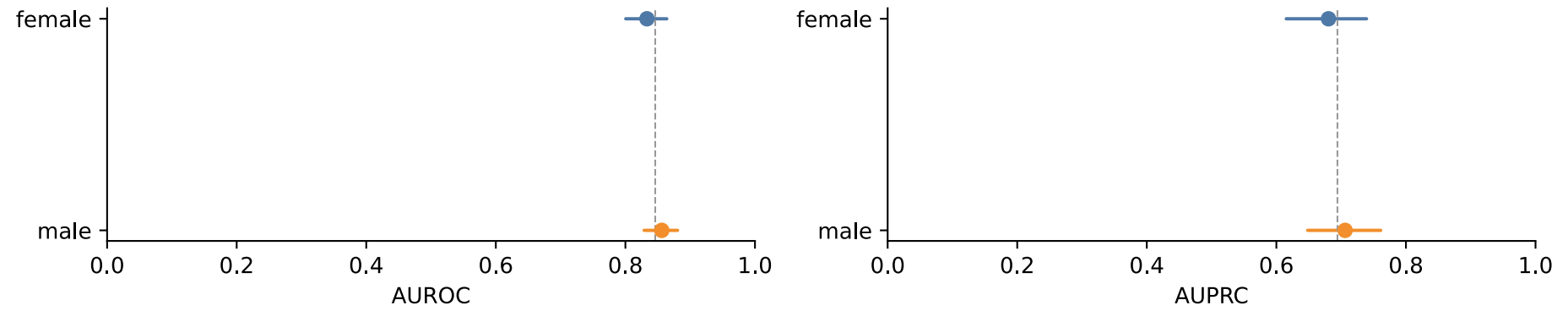
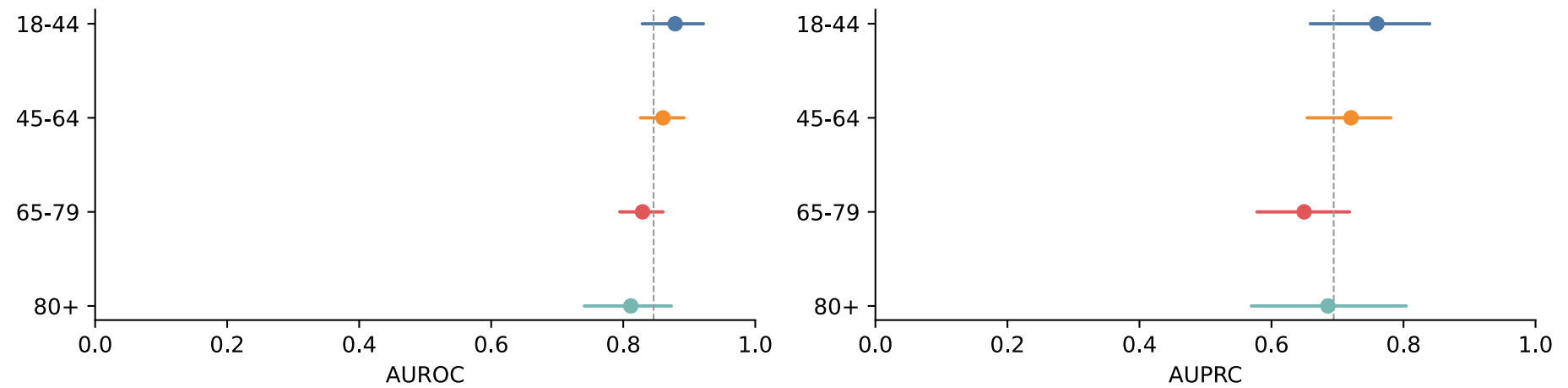
Discrimination

Attribute	Subgroup	n	Events	Prevalence	AUROC (95% CI)	AUPRC (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)
sex	Overall	1821	472	0.259	0.846 (0.825–0.866)	0.694 (0.650–0.734)	0.699 (0.657–0.741)	0.829 (0.808–0.848)
sex	female	787	199	0.253	0.833 (0.800–0.864)	0.680 (0.615–0.739)	0.724 (0.662–0.784)	0.811 (0.778–0.844)
sex	male	1034	273	0.264	0.856 (0.829–0.881)	0.706 (0.648–0.761)	0.681 (0.626–0.739)	0.842 (0.816–0.868)
age_group	Overall	1821	472	0.259	0.846 (0.825–0.866)	0.694 (0.650–0.734)	0.699 (0.657–0.741)	0.829 (0.808–0.848)
age_group	18-44	257	65	0.253	0.879 (0.829–0.922)	0.760 (0.659–0.840)	0.738 (0.623–0.836)	0.833 (0.777–0.883)
age_group	45-64	684	171	0.250	0.860 (0.826–0.892)	0.721 (0.654–0.781)	0.673 (0.602–0.742)	0.879 (0.851–0.907)
age_group	65-79	677	172	0.254	0.829 (0.795–0.860)	0.649 (0.578–0.718)	0.692 (0.622–0.758)	0.798 (0.760–0.832)
age_group	80+	203	64	0.315	0.811 (0.741–0.873)	0.686 (0.570–0.804)	0.750 (0.644–0.848)	0.748 (0.674–0.818)
race_ethnicity	Overall	1821	472	0.259	0.846 (0.825–0.866)	0.694 (0.650–0.734)	0.699 (0.657–0.741)	0.829 (0.808–0.848)
race_ethnicity	asian	13	5	0.385	0.800	0.630	0.600	0.750
race_ethnicity	black	266	82	0.308	0.809 (0.750–0.860)	0.691 (0.593–0.783)	0.683 (0.576–0.785)	0.821 (0.767–0.876)
race_ethnicity	hispanic	57	10	0.175	0.815	0.628	0.700	0.787
race_ethnicity	other	32	9	0.281	0.841	0.769	0.667	0.913
race_ethnicity	white	1453	366	0.252	0.855 (0.833–0.878)	0.697 (0.650–0.745)	0.705 (0.661–0.751)	0.831 (0.807–0.853)

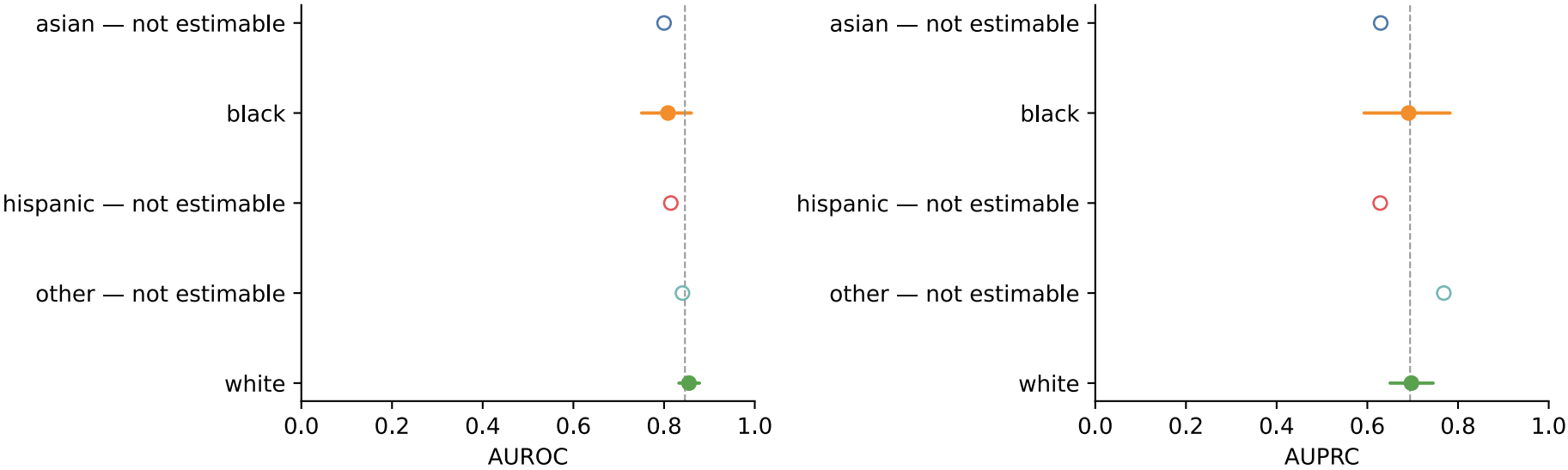
Gap statistics

Effect-size column shows Cohen's *h* on rate-metric gaps for two-group attributes; multi-group attributes and ranking metrics (AUROC / AUPRC) show "—".

Attribute	Metric	Value	Effect size
sex	equal_opportunity_difference	0.0423	h=+0.09 (negligible)
sex	predictive_equality	0.0311	h=-0.08 (negligible)
sex	equalized_odds_gap	0.0423	—
sex	statistical_parity	0.0281	—
age_group	equal_opportunity_difference	0.0775	—
age_group	predictive_equality	0.1309	—
age_group	equalized_odds_gap	0.1309	—
age_group	statistical_parity	0.1501	—
race_ethnicity	equal_opportunity_difference	0.1049	—
race_ethnicity	predictive_equality	0.1630	—
race_ethnicity	equalized_odds_gap	0.1630	—
race_ethnicity	statistical_parity	0.1346	—

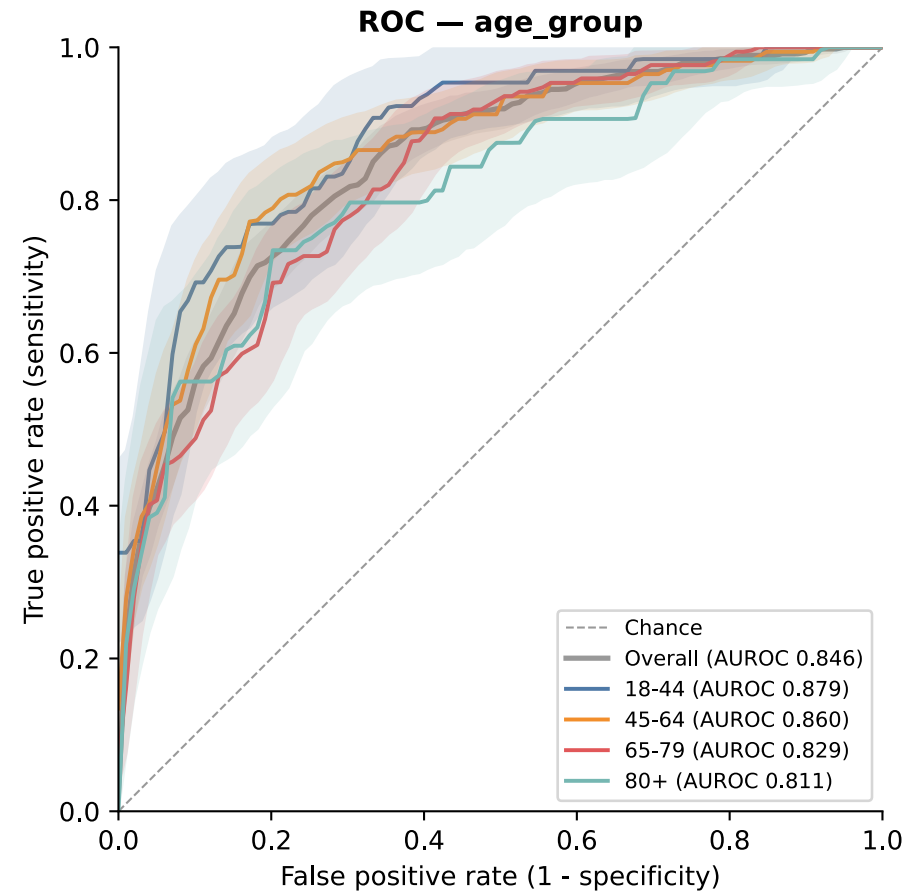
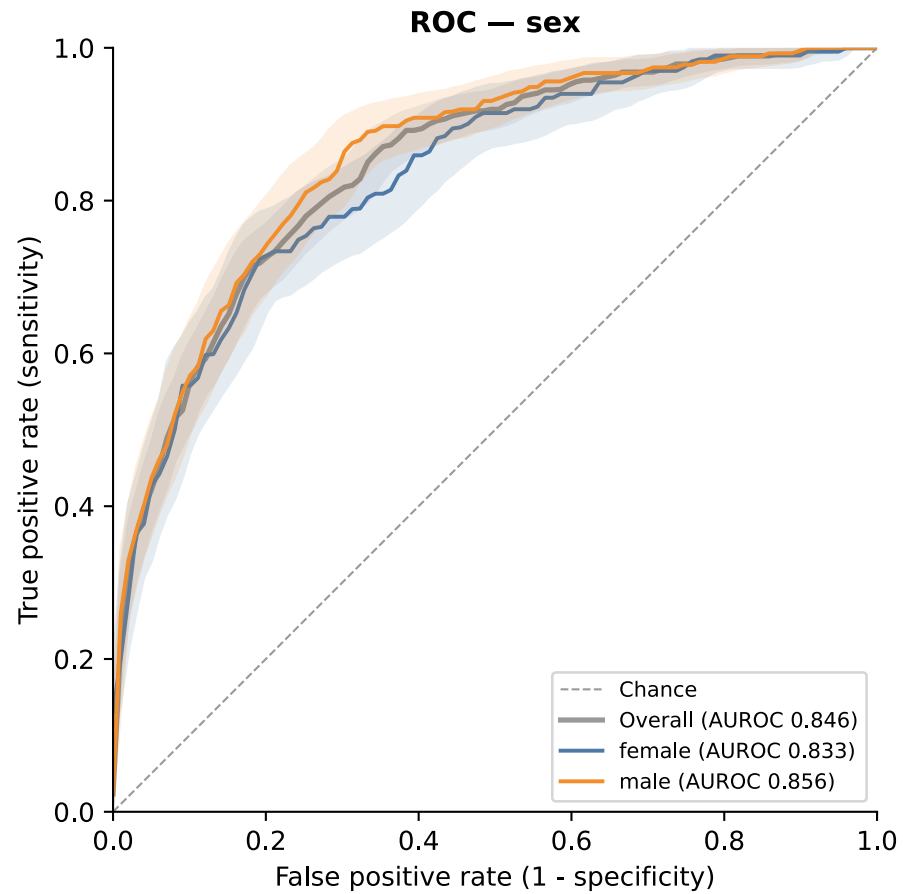
Discrimination — sex**Discrimination — age_group**

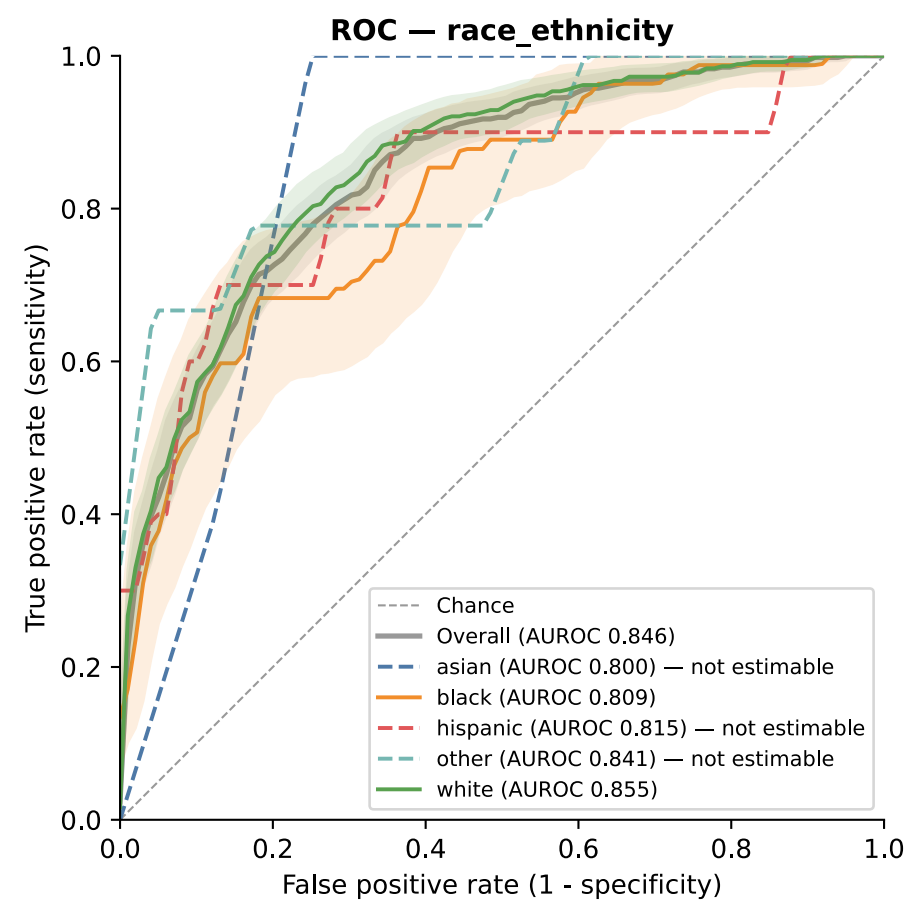
Discrimination — race_ethnicity



ROC curves

Per-subgroup ROC curves with pointwise 95% bootstrap bands. AUROC values in the legend are taken from the table above, not integrated from the drawn curve. Curves that cross indicate that no single subgroup dominates at every operating point — the ranking of subgroups depends on the threshold chosen.



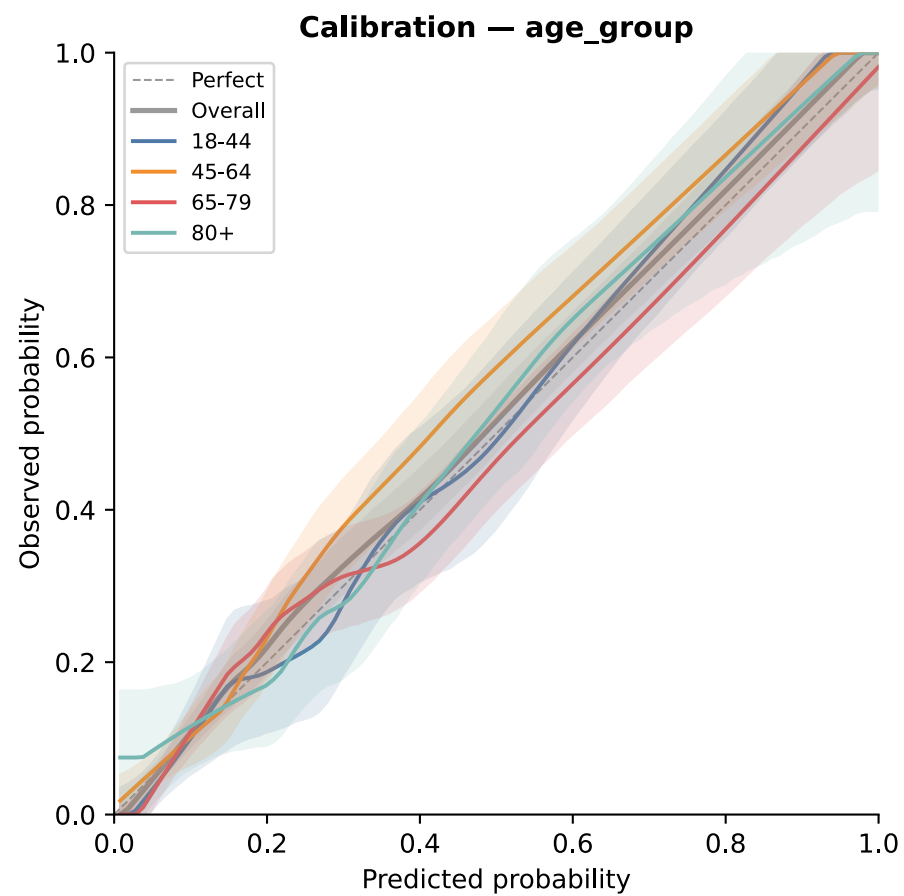
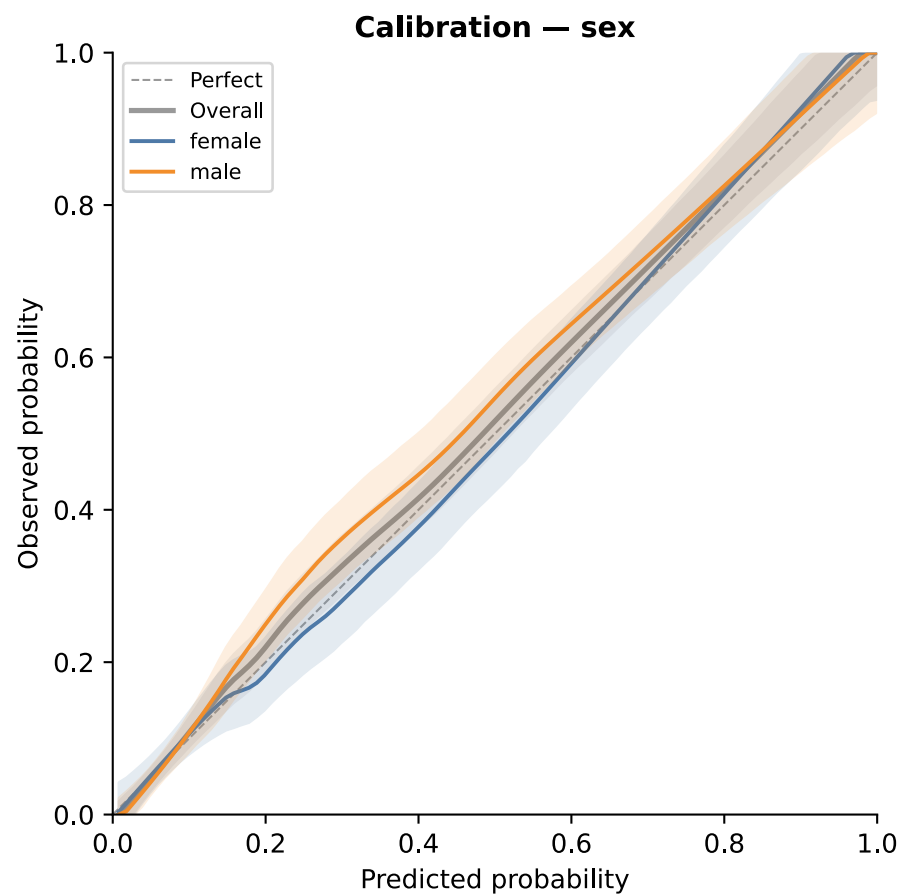


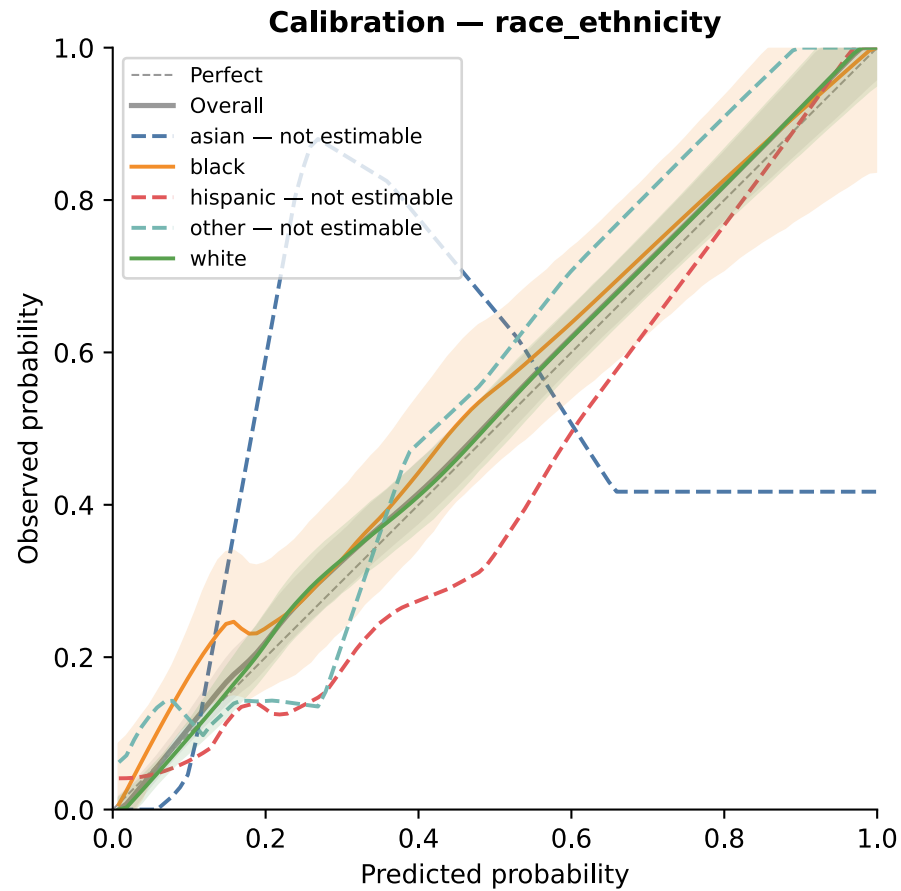
Calibration

Attribute	Subgroup	n	Intercept (95% CI)	Slope (95% CI)	Brier (95% CI)	ECE	ICI
sex	Overall	1821	0.082 (-0.048–0.205)	1.034 (0.931–1.146)	0.129 (0.120–0.139)	0.0191	0.0137
sex	female	787	-0.030 (-0.224–0.157)	1.008 (0.872–1.169)	0.131 (0.117–0.145)	0.0228	0.0095
sex	male	1034	0.171 (-0.002–0.338)	1.063 (0.925–1.224)	0.128 (0.115–0.141)	0.0344	0.0265
age_group	Overall	1821	0.082 (-0.048–0.205)	1.034 (0.931–1.146)	0.129 (0.120–0.139)	0.0191	0.0137
age_group	18-44	257	-0.025 (-0.380–0.295)	1.174	0.112 (0.090–0.135)	0.0333	0.0192
age_group	45-64	684	0.267 (0.056–0.463)	1.113 (0.941–1.335)	0.121 (0.106–0.138)	0.0380	0.0311
age_group	65-79	677	-0.049 (-0.270–0.159)	0.963 (0.821–1.129)	0.137 (0.121–0.153)	0.0357	0.0258
age_group	80+	203	0.078 (-0.282–0.401)	0.978	0.154 (0.126–0.186)	0.0387	0.0246
race_ethnicity	Overall	1821	0.082 (-0.048–0.205)	1.034 (0.931–1.146)	0.129 (0.120–0.139)	0.0191	0.0137
race_ethnicity	asian	13	0.675	0.949	0.201	0.3488	0.1884
race_ethnicity	black	266	0.336 (-0.018–0.654)	0.871	0.157 (0.130–0.186)	0.0552	0.0432
race_ethnicity	hispanic	57	-0.493	0.948	0.106	0.0981	0.0571
race_ethnicity	other	32	0.431	0.941	0.126	0.1236	0.0694
race_ethnicity	white	1453	0.039 (-0.111–0.181)	1.078 (0.970–1.218)	0.125 (0.114–0.135)	0.0238	0.0141

Calibration gap statistics

Attribute	Metric	Value (95% CI)
sex	max_intercept_gap	0.2009 (0.0120–0.4616)
sex	max_slope_gap	0.0547 (0.0036–0.2709)
age_group	max_intercept_gap	0.3165 (0.1663–0.7463)
age_group	max_slope_gap	0.2112
race_ethnicity	max_intercept_gap	1.1673
race_ethnicity	max_slope_gap	0.2070





Hosmer–Lemeshow goodness-of-fit

Read every statistic beside its n and its decile table. The Hosmer–Lemeshow statistic scales with subgroup size: a clinically irrelevant deviation becomes significant in a large subgroup, while a large deviation can go undetected in a small one. Values are therefore **not comparable across subgroups of different size**, and subgroups must not be ranked by p-value. This is why Kramer & Zimmerman (2007) require the statistic to be reported together with the number of patients, the per-decile observed-versus-expected table below, and the adjunct calibration measures in the table above — never on its own.

Attribute	Subgroup	n	Events	H–L statistic	df	p	Groups used	Decile slope	Decile intercept
sex	Overall	1821	472	6.98	10	0.7269	10	1.035	+0.002
sex	female	787	199	4.89	10	0.8984	10	1.000	-0.004
sex	male	1034	273	11.61	10	0.3120	10	1.074	+0.004
age_group	Overall	1821	472	6.98	10	0.7269	10	1.035	+0.002

Attribute	Subgroup	n	Events	H–L statistic	df	p	Groups used	Decile slope	Decile intercept
age_group	18-44	257	65	3.46	10	0.9686	10	1.060	-0.018
age_group	45-64	684	171	9.86	10	0.4526	10	1.114	+0.008
age_group	65-79	677	172	9.41	10	0.4934	10	0.928	+0.012
age_group	80+	203	64	2.58	10	0.9896	10	1.031	+0.003
race_ethnicity	Overall	1821	472	6.98	10	0.7269	10	1.035	+0.002
race_ethnicity	asian	13	5	12.59	10	0.2475	10	0.650	+0.277
race_ethnicity	black	266	82	11.80	10	0.2984	10	0.953	+0.058
race_ethnicity	hispanic	57	10	7.16	10	0.7099	10	0.848	-0.020
race_ethnicity	other	32	9	10.85	10	0.3693	10	1.061	+0.040
race_ethnicity	white	1453	366	7.20	10	0.7063	10	1.046	-0.006

† Fewer risk groups than requested were usable, because the subgroup was small or its predicted probabilities were tied. Degrees of freedom follow the reduced count. A well-calibrated model gives a decile slope near 1 and a decile intercept near 0 (Kramer & Zimmerman 2007). 14 tests are reported here; no multiplicity adjustment has been applied. Adjust within an attribute if needed — do not pool the “Overall” rows with the subgroup rows.

Attribute	Metric	Value
sex	max_hl_gap	6.72
age_group	max_hl_gap	7.28
race_ethnicity	max_hl_gap	5.43

The gap is the largest pairwise difference in the statistic across subgroups. It inherits the sample-size dependence in full and should be read as a descriptive summary, not as evidence of differential calibration.

Observed versus expected events per decile

Deciles of predicted risk, of approximately equal size, with cut points snapped to tied predicted probabilities so that identical predictions always share a decile.

sex — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	183	4	4.5	0.0219	0.0248	-0.5
2	182	6	7.7	0.0330	0.0425	-1.7
3	182	11	10.9	0.0604	0.0601	+0.1
4	182	17	15.4	0.0934	0.0848	+1.6
5	182	20	22.8	0.1099	0.1253	-2.8
6	182	42	33.3	0.2308	0.1827	+8.7
7	182	50	47.4	0.2747	0.2605	+2.6
8	182	76	68.1	0.4176	0.3741	+7.9
9	182	94	96.6	0.5165	0.5308	-2.6
10	182	152	145.8	0.8352	0.8012	+6.2

age_group — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	183	4	4.5	0.0219	0.0248	-0.5
2	182	6	7.7	0.0330	0.0425	-1.7
3	182	11	10.9	0.0604	0.0601	+0.1
4	182	17	15.4	0.0934	0.0848	+1.6
5	182	20	22.8	0.1099	0.1253	-2.8
6	182	42	33.3	0.2308	0.1827	+8.7
7	182	50	47.4	0.2747	0.2605	+2.6
8	182	76	68.1	0.4176	0.3741	+7.9
9	182	94	96.6	0.5165	0.5308	-2.6
10	182	152	145.8	0.8352	0.8012	+6.2

sex — female

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	79	2	2.0	0.0253	0.0256	-0.0
2	79	3	3.5	0.0380	0.0447	-0.5
3	79	7	5.1	0.0886	0.0647	+1.9
4	79	5	7.3	0.0633	0.0925	-2.3
5	79	15	11.1	0.1899	0.1406	+3.9
6	79	15	15.6	0.1899	0.1974	-0.6
7	79	18	22.0	0.2278	0.2780	-4.0
8	78	28	31.1	0.3590	0.3986	-3.1
9	78	43	42.7	0.5513	0.5473	+0.3
10	78	63	61.7	0.8077	0.7909	+1.3

age_group — 18-44

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	26	0	0.5	0.0000	0.0198	-0.5
2	26	1	0.9	0.0385	0.0349	+0.1
3	26	1	1.4	0.0385	0.0535	-0.4
4	26	1	2.3	0.0385	0.0869	-1.3
5	26	3	3.3	0.1154	0.1274	-0.3
6	26	6	4.7	0.2308	0.1807	+1.3
7	26	5	7.1	0.1923	0.2749	-2.1
8	25	12	10.3	0.4800	0.4123	+1.7
9	25	14	14.1	0.5600	0.5622	-0.1
10	25	22	21.2	0.8800	0.8481	+0.8

sex — male

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	104	2	2.5	0.0192	0.0243	-0.5
2	104	5	4.3	0.0481	0.0411	+0.7
3	104	4	6.0	0.0385	0.0573	-2.0
4	104	9	8.4	0.0865	0.0804	+0.6
5	103	8	12.1	0.0777	0.1172	-4.1
6	103	24	17.6	0.2330	0.1712	+6.4
7	103	33	25.8	0.3204	0.2501	+7.2
8	103	44	37.1	0.4272	0.3599	+6.9
9	103	56	53.4	0.5437	0.5182	+2.6
10	103	88	83.5	0.8544	0.8104	+4.5

age_group — 45-64

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	69	1	1.4	0.0145	0.0209	-0.4
2	69	3	2.3	0.0435	0.0327	+0.7
3	69	4	3.2	0.0580	0.0462	+0.8
4	69	7	4.5	0.1014	0.0653	+2.5
5	68	5	6.5	0.0735	0.0951	-1.5
6	68	11	10.0	0.1618	0.1469	+1.0
7	68	18	14.8	0.2647	0.2179	+3.2
8	68	29	22.0	0.4265	0.3231	+7.0
9	68	35	31.5	0.5147	0.4639	+3.5
10	68	58	52.7	0.8529	0.7746	+5.3

age_group — 65-79

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	68	0	2.3	0.0000	0.0338	-2.3
2	68	4	3.5	0.0588	0.0512	+0.5
3	68	3	4.8	0.0441	0.0700	-1.8
4	68	6	6.5	0.0882	0.0958	-0.5
5	68	13	9.5	0.1912	0.1394	+3.5
6	68	18	13.3	0.2647	0.1962	+4.7
7	68	20	18.8	0.2941	0.2767	+1.2
8	67	24	26.4	0.3582	0.3938	-2.4
9	67	32	37.7	0.4776	0.5623	-5.7
10	67	52	53.7	0.7761	0.8009	-1.7

age_group — 80+

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	21	1	1.0	0.0476	0.0487	-0.0
2	21	2	1.5	0.0952	0.0691	+0.5
3	21	3	2.1	0.1429	0.0983	+0.9
4	20	4	3.0	0.2000	0.1490	+1.0
5	20	3	4.2	0.1500	0.2076	-1.2
6	20	4	5.4	0.2000	0.2699	-1.4
7	20	8	7.1	0.4000	0.3539	+0.9
8	20	9	9.1	0.4500	0.4565	-0.1
9	20	13	11.9	0.6500	0.5925	+1.1
10	20	17	16.4	0.8500	0.8211	+0.6

race_ethnicity — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	183	4	4.5	0.0219	0.0248	-0.5
2	182	6	7.7	0.0330	0.0425	-1.7
3	182	11	10.9	0.0604	0.0601	+0.1
4	182	17	15.4	0.0934	0.0848	+1.6
5	182	20	22.8	0.1099	0.1253	-2.8
6	182	42	33.3	0.2308	0.1827	+8.7
7	182	50	47.4	0.2747	0.2605	+2.6
8	182	76	68.1	0.4176	0.3741	+7.9
9	182	94	96.6	0.5165	0.5308	-2.6
10	182	152	145.8	0.8352	0.8012	+6.2

race_ethnicity — asian

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	2	0	0.1	0.0000	0.0333	-0.1
2	2	0	0.1	0.0000	0.0656	-0.1
3	2	0	0.2	0.0000	0.0977	-0.2
4	1	1	0.2	1.0000	0.2409	+0.8
5	1	1	0.3	1.0000	0.2629	+0.7
6	1	1	0.4	1.0000	0.3610	+0.6
7	1	0	0.5	0.0000	0.5318	-0.5
8	1	1	0.6	1.0000	0.5672	+0.4
9	1	1	0.6	1.0000	0.6163	+0.4
10	1	0	0.7	0.0000	0.6575	-0.7

race_ethnicity — black

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	27	1	0.6	0.0370	0.0227	+0.4
2	27	1	1.2	0.0370	0.0439	-0.2
3	27	4	1.8	0.1481	0.0652	+2.2
4	27	4	2.7	0.1481	0.0990	+1.3
5	27	8	3.8	0.2963	0.1419	+4.2
6	27	8	5.6	0.2963	0.2065	+2.4
7	26	7	7.6	0.2692	0.2939	-0.6
8	26	11	10.6	0.4231	0.4063	+0.4
9	26	17	14.5	0.6538	0.5564	+2.5
10	26	21	21.3	0.8077	0.8208	-0.3

race_ethnicity — hispanic

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	6	0	0.1	0.0000	0.0211	-0.1
2	6	1	0.2	0.1667	0.0354	+0.8
3	6	0	0.3	0.0000	0.0557	-0.3
4	6	0	0.5	0.0000	0.0830	-0.5
5	6	0	0.8	0.0000	0.1304	-0.8
6	6	1	1.1	0.1667	0.1907	-0.1
7	6	1	1.7	0.1667	0.2830	-0.7
8	5	1	1.9	0.2000	0.3772	-0.9
9	5	3	2.6	0.6000	0.5154	+0.4
10	5	3	3.9	0.6000	0.7827	-0.9

race_ethnicity — other

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	4	0	0.1	0.0000	0.0162	-0.1
2	4	0	0.1	0.0000	0.0315	-0.1
3	3	1	0.1	0.3333	0.0493	+0.9
4	3	1	0.2	0.3333	0.0654	+0.8
5	3	0	0.3	0.0000	0.1055	-0.3
6	3	0	0.5	0.0000	0.1775	-0.5
7	3	1	0.6	0.3333	0.2041	+0.4
8	3	1	1.1	0.3333	0.3797	-0.1
9	3	2	1.7	0.6667	0.5667	+0.3
10	3	3	2.6	1.0000	0.8560	+0.4

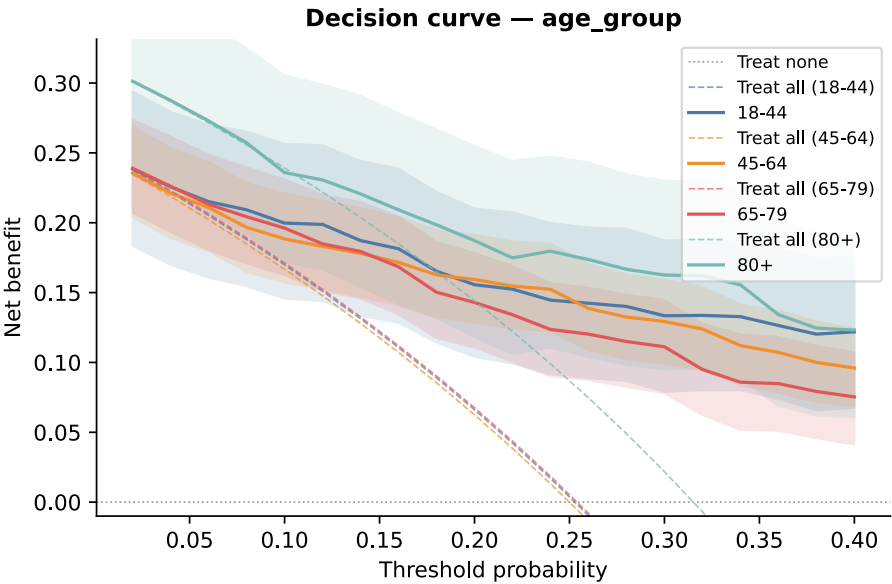
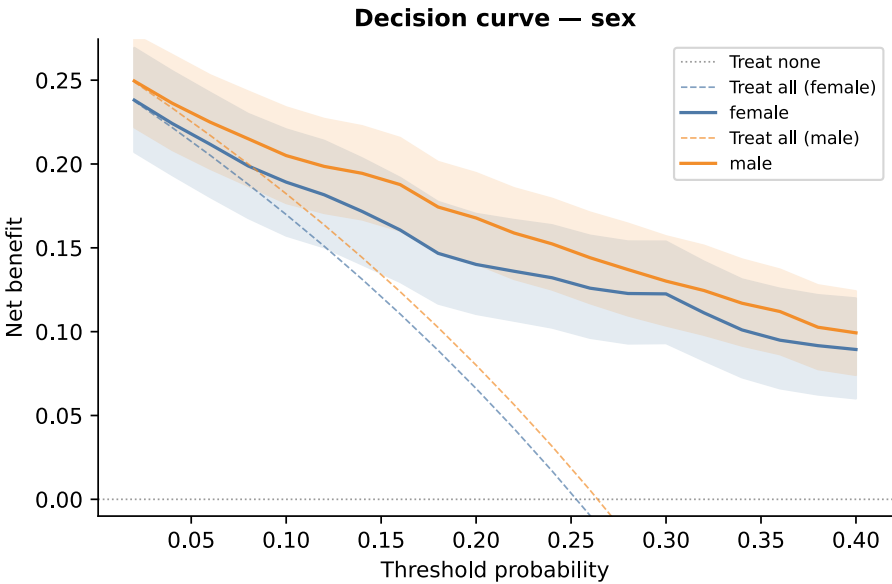
race_ethnicity — white

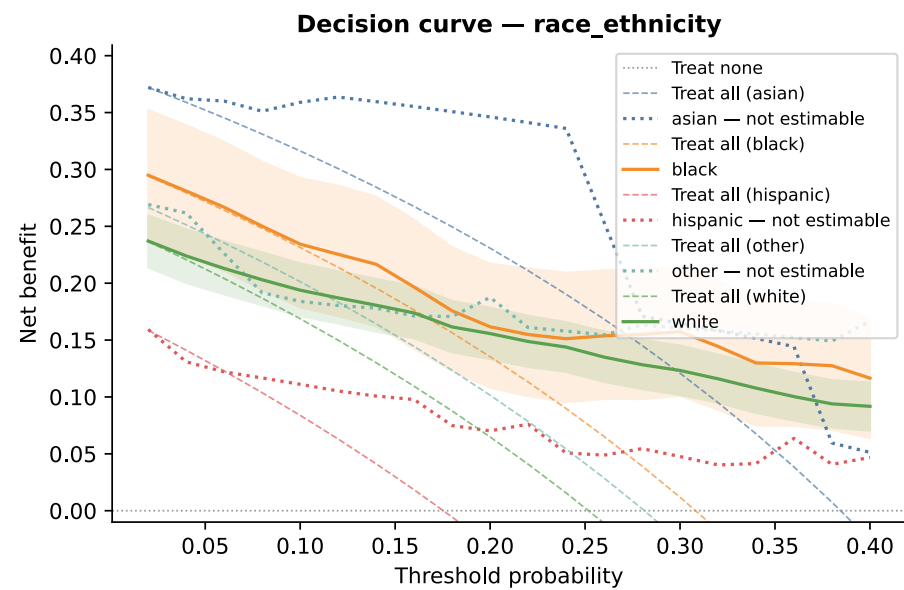
Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	146	3	3.8	0.0205	0.0259	-0.8
2	146	5	6.3	0.0342	0.0430	-1.3
3	146	6	8.8	0.0411	0.0603	-2.8
4	145	12	12.2	0.0828	0.0840	-0.2
5	145	15	18.1	0.1034	0.1245	-3.1
6	145	30	26.2	0.2069	0.1806	+3.8
7	145	41	37.4	0.2828	0.2578	+3.6
8	145	61	53.6	0.4207	0.3697	+7.4
9	145	71	76.6	0.4897	0.5282	-5.6
10	145	122	115.9	0.8414	0.7994	+6.1

Clinical utility (decision curves)

Clinical utility gap statistics

Attribute	Metric	Value (95% CI)
sex	max_net_benefit_gap	0.0277 (0.0131–0.0719)
age_group	max_net_benefit_gap	0.0697 (0.0501–0.1662)
race_ethnicity	max_net_benefit_gap	0.2852





Mitigation comparison

sex

Calibration

Recalibration targets calibration, so this is where its effect belongs. Compare the intercept against 0 and the slope against 1 within each subgroup, and the gap statistics below across subgroups.

Before

Subgroup	Intercept (95% CI)	Slope (95% CI)	Brier (95% CI)	ECE	ICI
female	-0.030 (-0.224–0.157)	1.008 (0.872–1.169)	0.131 (0.117–0.145)	0.0228	0.0095
male	0.171 (-0.002–0.338)	1.063 (0.925–1.224)	0.128 (0.115–0.141)	0.0344	0.0265

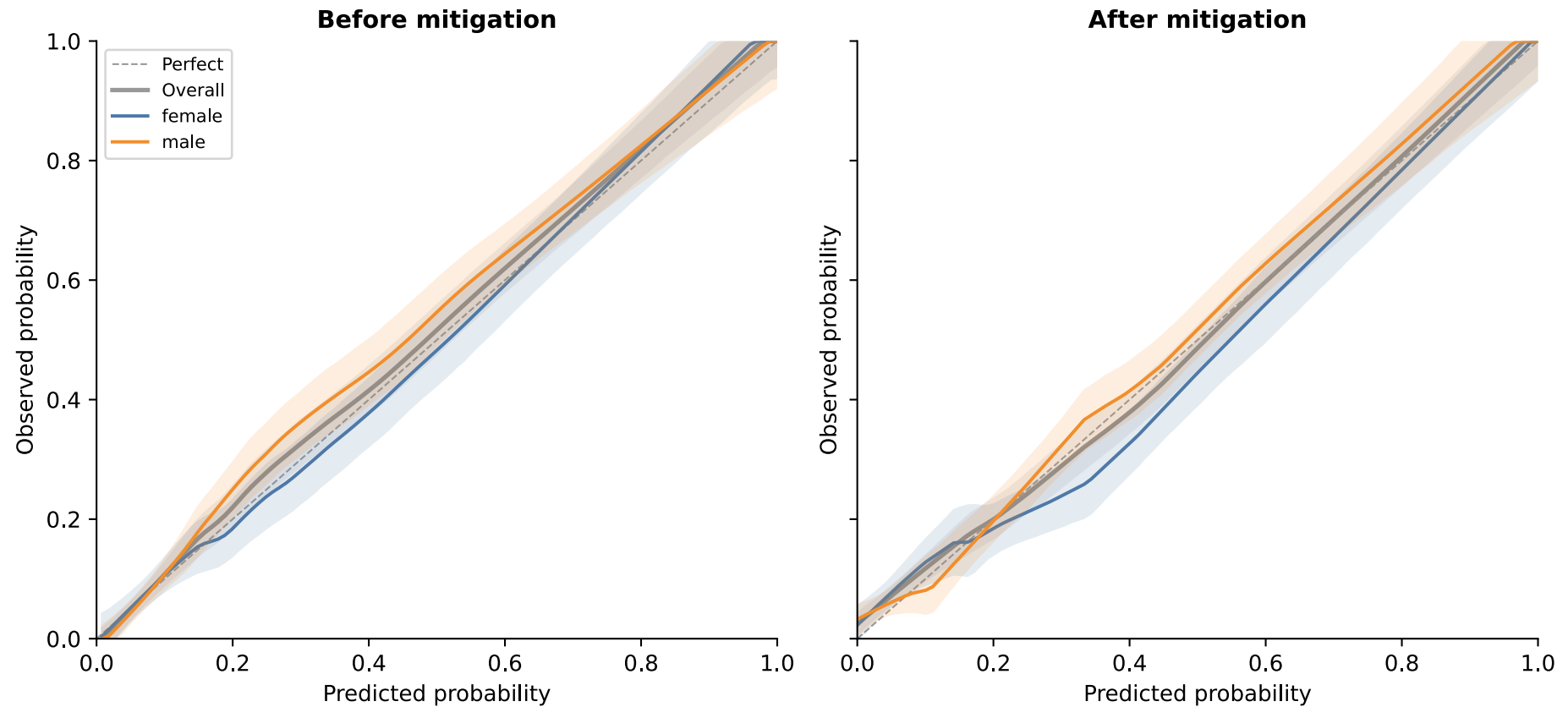
After

Subgroup	Intercept (95% CI)	Slope (95% CI)	Brier (95% CI)	ECE	ICI
female	-0.081 (-0.288–0.118)	0.569 (0.257–0.904)	0.133 (0.119–0.148)	0.0442	0.0385
male	0.153 (-0.031–0.328)	0.827 (0.609–1.043)	0.129 (0.116–0.142)	0.0389	0.0211

Calibration gaps

Metric	Before (95% CI)	After (95% CI)
max_intercept_gap	0.2009 (0.0120–0.4616)	0.2339 (0.0178–0.5112)
max_slope_gap	0.0547 (0.0036–0.2709)	0.2577 (0.0153–0.6313)

Calibration before and after — sex



Discrimination (sanity check)

Per-subgroup recalibration is a monotone transform within each subgroup, so within-subgroup AUROC is invariant except for ties that isotonic regression introduces, and any sensitivity change is a threshold artefact. These columns barely moving is the expected result, not a null finding — a large shift here means the mitigation did something other than recalibrate.

Before

Subgroup	AUROC	Sensitivity
female	0.833 (0.800–0.864)	0.724 (0.662–0.784)
male	0.856 (0.829–0.881)	0.681 (0.626–0.739)

After

Subgroup	AUROC	Sensitivity
female	0.825 (0.793–0.857)	0.779 (0.721–0.833)
male	0.853 (0.827–0.877)	0.722 (0.668–0.774)

age_group

Calibration

Recalibration targets calibration, so this is where its effect belongs. Compare the intercept against 0 and the slope against 1 within each subgroup, and the gap statistics below across subgroups.

Before

Subgroup	Intercept (95% CI)	Slope (95% CI)	Brier (95% CI)	ECE	ICI
18-44	-0.025 (-0.380–0.295)	1.174	0.112 (0.090–0.135)	0.0333	0.0192
45-64	0.267 (0.056–0.463)	1.113 (0.941–1.335)	0.121 (0.106–0.138)	0.0380	0.0311
65-79	-0.049 (-0.270–0.159)	0.963 (0.821–1.129)	0.137 (0.121–0.153)	0.0357	0.0258
80+	0.078 (-0.282–0.401)	0.978	0.154 (0.126–0.186)	0.0387	0.0246

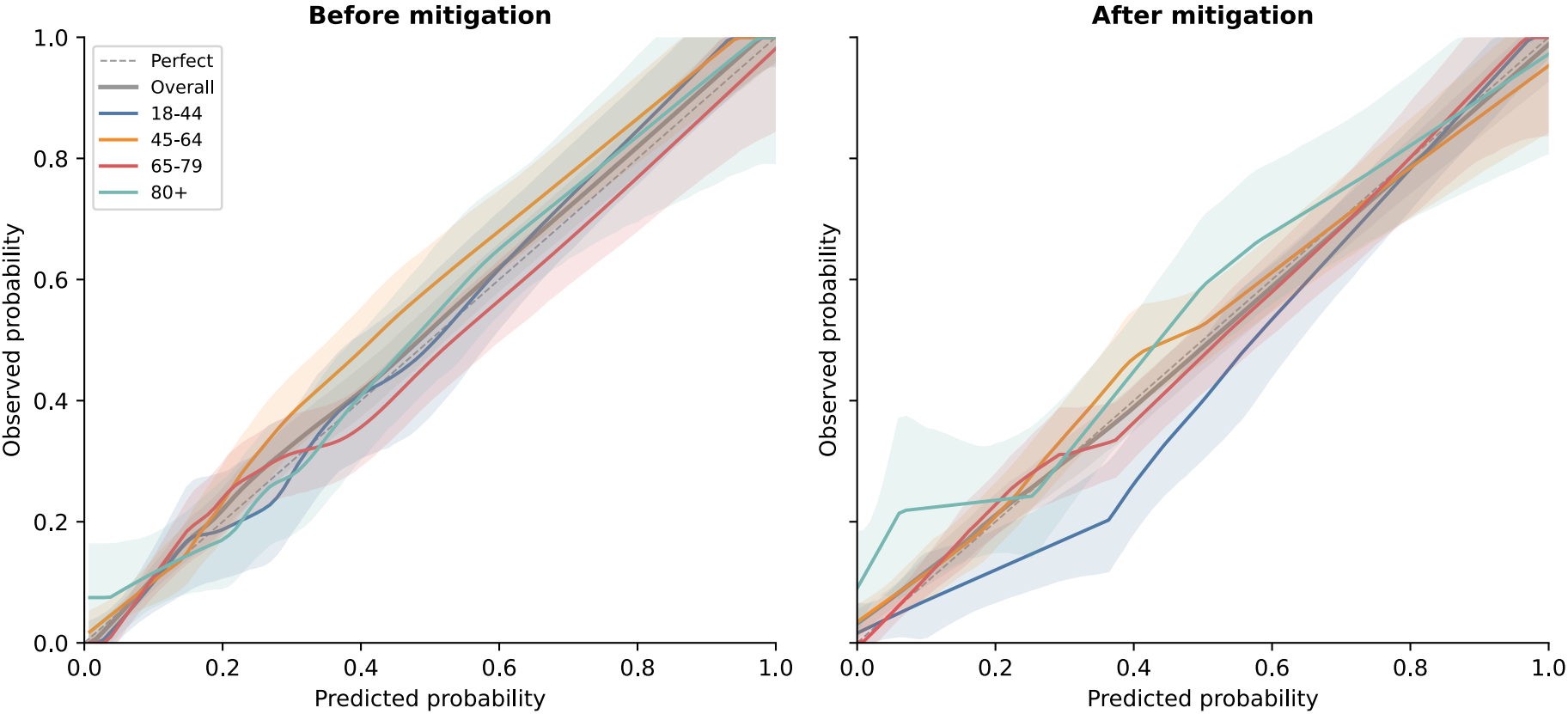
After

Subgroup	Intercept (95% CI)	Slope (95% CI)	Brier (95% CI)	ECE	ICI
18-44	-0.421 (-0.814–0.071)	0.557	0.115 (0.096–0.135)	0.0709	0.0593
45-64	0.220 (-0.017–0.450)	0.327 (0.150–0.732)	0.123 (0.107–0.141)	0.0328	0.0294
65-79	-0.006 (-0.227–0.201)	0.939 (0.782–1.117)	0.139 (0.124–0.156)	0.0450	0.0167
80+	0.592 (0.135–1.037)	0.099	0.160 (0.127–0.198)	0.0960	0.0711

Calibration gaps

Metric	Before (95% CI)	After (95% CI)
max_intercept_gap	0.3165 (0.1663–0.7463)	1.0128 (0.5203–1.5683)
max_slope_gap	0.2112	0.8393

Calibration before and after — age_group



Discrimination (sanity check)

Per-subgroup recalibration is a monotone transform within each subgroup, so within-subgroup AUROC is invariant except for ties that isotonic regression introduces, and any sensitivity change is a threshold artefact. These columns barely moving is the expected result, not a null finding — a large shift here means the mitigation did something other than recalibrate.

Before

Subgroup	AUROC	Sensitivity
18-44	0.879 (0.829–0.922)	0.738 (0.623–0.836)
45-64	0.860 (0.826–0.892)	0.673 (0.602–0.742)

After

Subgroup	AUROC	Sensitivity
18-44	0.881 (0.832–0.923)	0.923 (0.855–0.983)
45-64	0.855 (0.821–0.888)	0.696 (0.626–0.765)

Subgroup	AUROC	Sensitivity
65-79	0.829 (0.795–0.860)	0.692 (0.622–0.758)
80+	0.811 (0.741–0.873)	0.750 (0.644–0.848)

Subgroup	AUROC	Sensitivity
65-79	0.827 (0.793–0.859)	0.605 (0.530–0.677)
80+	0.795 (0.724–0.858)	0.562 (0.441–0.677)

race_ethnicity

Calibration

Recalibration targets calibration, so this is where its effect belongs. Compare the intercept against 0 and the slope against 1 within each subgroup, and the gap statistics below across subgroups.

Before

Subgroup	Intercept (95% CI)	Slope (95% CI)	Brier (95% CI)	ECE	ICI
asian	0.675	0.949	0.201	0.3488	0.1884
black	0.336 (-0.018–0.654)	0.871	0.157 (0.130–0.186)	0.0552	0.0432
hispanic	-0.493	0.948	0.106	0.0981	0.0571
other	0.431	0.941	0.126	0.1236	0.0694
white	0.039 (-0.111–0.181)	1.078 (0.970–1.218)	0.125 (0.114–0.135)	0.0238	0.0141

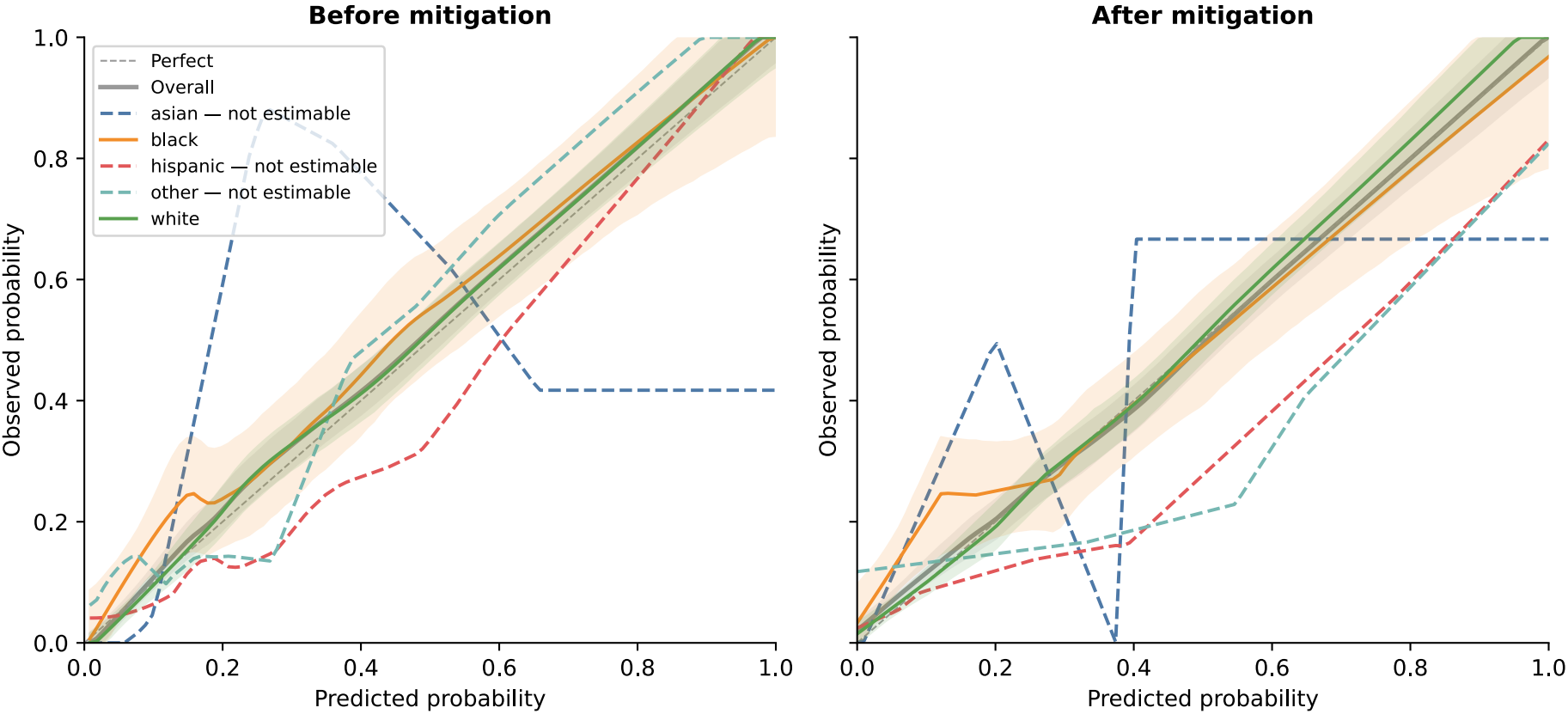
After

Subgroup	Intercept (95% CI)	Slope (95% CI)	Brier (95% CI)	ECE	ICI
asian	0.998	0.647	0.235	0.3833	0.2295
black	0.247 (-0.133–0.620)	0.443	0.159 (0.131–0.188)	0.0627	0.0482
hispanic	-0.980	0.114	0.123	0.1315	0.1058
other	-0.732	0.079	0.147	0.1623	0.1496
white	0.052 (-0.102–0.201)	0.906 (0.725–1.102)	0.125 (0.115–0.136)	0.0278	0.0101

Calibration gaps

Metric	Before (95% CI)	After (95% CI)
max_intercept_gap	1.1673	1.9782
max_slope_gap	0.2070	0.8265

Calibration before and after — race_ethnicity



Discrimination (sanity check)

Per-subgroup recalibration is a monotone transform within each subgroup, so within-subgroup AUROC is invariant except for ties that isotonic regression introduces, and any sensitivity change is a threshold artefact. These columns barely moving is the expected result, not a null finding — a large shift here means the mitigation did something other than recalibrate.

Before

Subgroup	AUROC	Sensitivity
asian	0.800	0.600
black	0.809 (0.750–0.860)	0.683 (0.576–0.785)

After

Subgroup	AUROC	Sensitivity
asian	0.712	0.400
black	0.804 (0.746–0.856)	0.659 (0.553–0.760)

Subgroup	AUROC	Sensitivity
hispanic	0.815	0.700
other	0.841	0.667
white	0.855 (0.833–0.878)	0.705 (0.661–0.751)

Subgroup	AUROC	Sensitivity
hispanic	0.818	0.800
other	0.809	0.778
white	0.853 (0.831–0.876)	0.740 (0.694–0.787)

Appendix: TRIPOD+AI item coverage

Fairness-relevant items from the TRIPOD+AI checklist (Collins et al., 2024, *BMJ*). Items addressed by this audit are linked to the relevant report section. Items marked "User responsibility" require information that the audit tool cannot produce automatically.

Item	Description	Status	Detail
3c	Describe any known health inequalities between sociodemographic groups	User responsibility	User must provide context in manuscript introduction
5a	Describe data sources, rationale, and representativeness	User responsibility	User must describe data representativeness in manuscript methods
7	Describe data pre-processing, including whether similar across sociodemographic groups	User responsibility	User must verify preprocessing does not vary by group
8a	Outcome assessment consistency across sociodemographic groups	User responsibility	User must verify outcome labels are measured consistently
8b	Assessor demographics for outcome assessment	User responsibility	User must report assessor demographics
9c	Assessor demographics for predictor measurement	User responsibility	User must report assessor demographics
12e	All measures used to evaluate model performance	Addressed	Discrimination, Calibration, and Clinical utility sections
12f	Describe any model updating (eg recalibration) for sociodemographic groups or settings	Addressed	Mitigation section
14	Fairness approaches and rationale	Addressed	Audit summary and gap statistics throughout
20b	Report demographic characteristics and group differences	Addressed	Discrimination section (subgroup sizes, prevalences)
20c	Compare predictor distributions including demographics	User responsibility	User must report predictor distributions per group in manuscript
23a	Report performance estimates with CIs, including for key sociodemographic subgroups	Addressed	Discrimination, Calibration, and Clinical utility sections
25	Overall interpretation including fairness issues	User responsibility	User must interpret results in manuscript discussion
26	Limitations including non-representative sample, bias effects	User responsibility	User must discuss limitations in manuscript

Source: Collins GS, Moons KGM, Dhiman P, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* 2024;385:e078378. doi: [10.1136/bmj-2023-078378](https://doi.org/10.1136/bmj-2023-078378). This checklist covers fairness-relevant items only; see the full statement for the complete checklist.

Appendix: How to read this report

Discrimination

Discrimination measures how well the model separates positive from negative cases.

AUROC (Area Under the ROC Curve)

Probability that the model ranks a random positive case higher than a random negative case. Range [0, 1]; 0.5 = no discrimination; 1.0 = perfect separation.

AUPRC (Area Under the Precision-Recall Curve)

Summarises precision across all recall levels. More informative than AUROC for low-prevalence outcomes. Higher is better; baseline equals the prevalence.

Sensitivity (True Positive Rate)

Proportion of true positive cases correctly identified at the chosen threshold. = $TP / (TP + FN)$.

Specificity (True Negative Rate)

Proportion of true negative cases correctly identified at the chosen threshold. = $TN / (TN + FP)$.

PPV (Positive Predictive Value)

Proportion of positive predictions that are truly positive. = $TP / (TP + FP)$.

NPV (Negative Predictive Value)

Proportion of negative predictions that are truly negative. = $TN / (TN + FN)$.

Calibration

Calibration measures whether predicted probabilities match observed event rates.

Calibration intercept

Intercept from logistic recalibration with slope fixed to 1 (Steyerberg 2009). Perfect calibration = 0. Positive values indicate the model underestimates risk on average; negative values indicate overestimation.

Calibration slope

Slope from logistic recalibration with free intercept and slope. Perfect calibration = 1. Values < 1 indicate overconfident predictions (predicted probabilities too extreme); values > 1 indicate underconfident predictions.

Brier score

Mean squared difference between predicted probabilities and binary outcomes. Range [0, 1]; lower is better.

ECE (Expected Calibration Error)

Weighted average of absolute calibration error across equal-frequency bins. Lower is better.

ICI (Integrated Calibration Index)

Mean absolute difference between LOESS-smoothed observed probabilities and predicted probabilities (Austin & Steyerberg 2019). Lower is better; more robust than ECE as it avoids binning artefacts.

Hosmer–Lemeshow statistic

Chi-square-distributed goodness-of-fit statistic comparing observed with expected events across deciles of predicted risk. A large value indicates lack of fit. **It scales with sample size**, so it is not comparable between subgroups of different size and must be read alongside n , the per-decile table, and the calibration measures above (Kramer & Zimmerman 2007).

Hosmer–Lemeshow degrees of freedom

Equal to the number of risk groups used. The classical convention subtracts 2, which is the penalty for estimating an intercept and slope on the data under test; this package evaluates predictions supplied by a model fitted elsewhere, so no degrees of freedom are spent and none are subtracted.

Decile slope and intercept

Least-squares regression of the observed on the expected event rate across risk groups, recommended by Kramer & Zimmerman (2007) as an adjunct to the statistic. Perfect calibration gives a slope of 1 and an intercept of 0.

Clinical utility (decision curves)

Decision curve analysis quantifies the clinical value of using the model at a given decision threshold.

Net benefit

$TP/n - FP/n \times p_t / (1 - p_t)$, where p_t is the decision threshold. Higher is better. A model is clinically useful when its net benefit exceeds both 0 (treat none) and the treat-all reference.

Standardised net benefit

Net benefit divided by prevalence (Naderalvojud 2025). Normalised to a [0, 1] scale for comparison across cohorts with different prevalences.

Treat-all reference

Net benefit if all patients receive treatment. = prevalence – (1 – prevalence) $\times p_t / (1 - p_t)$. Becomes strongly negative at high thresholds.

Treat-none reference

Net benefit = 0. The baseline of no intervention.

Gap statistics

Gap statistics quantify the largest disparity between subgroups for each metric. All gaps are non-negative; 0 = perfect parity.

Equal opportunity difference

max – min of sensitivity across subgroups. Measures fairness in true positive identification.

Predictive equality

max – min of false positive rate (1 – specificity) across subgroups. Measures fairness in false alarm rates.

Equalized odds gap

Maximum of equal opportunity difference and predictive equality.

Statistical parity

max – min of positive prediction rate across subgroups. Measures whether the model predicts positive at equal rates.

Max intercept / slope gap

Maximum pairwise difference in calibration intercept or slope across subgroups, with bootstrap confidence intervals.

Max net benefit gap

Maximum across all thresholds of the maximum pairwise net-benefit difference between subgroups, with bootstrap confidence intervals.

Confidence intervals and shrinkage**Bootstrap confidence intervals**

95% percentile intervals from subgroup-stratified patient-level resamples. Patients are resampled with replacement within each subgroup, preserving the subgroup's size while letting its event count vary.

Empirical Bayes shrinkage

Interaction cells with fewer events than the configured threshold (shown in *italics*) use Beta-Binomial shrinkage toward the marginal subgroup rate. This reduces instability in small cells. Confidence intervals are suppressed for shrunk estimates.

Generated by **isitfair**. Three-axis evaluation: discrimination, calibration, and clinical utility reported jointly per TRIPOD+AI recommendations.

Bootstrap confidence intervals (95%) from 2000 subgroup-stratified patient-level resamples.